A survey of Mobility management for mobile networks supporting LEO satellite access

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***Abstract*— The integration of space, aerial and terrestrial access is an important feature of the future 6G network, and it is also one of the development trends of the current mobile network. In particular, Low Earth Orbit (LEO) satellite has the advantages of high bandwidth, low delay and wide coverage, which can be a powerful supplement to the current mobile access method. However, the ultra-high mobility of LEO satellite brings severe challenges to the mobility management of legacy mobile networks. This paper first introduces the mobile network architecture supporting satellite access, then presents the main mobility management methods in the current IP network, and analyzes the pros and cons of various methods respectively. Then, we makes a comprehensive analysis on the mobility management methods of 5G mobile network, and puts forward the optimization idea of mobility management of satellite access, After that, we analyzes the two mobility schemes of satellite access proposed by 3GPP in SAT_ARCH project. Finally, the future research direction of mobility management of subsequent satellite access mobile network are discussed.**

Keywords—LEO, Mobility Management, NTN

I. INTRODUCTION

The number of global mobile users has reached 5.3 billion and the number of mobile connections has exceeded 10 billion [1] by Q2 of 2022. However, the existing mobile networks are mainly based on ground base stations. For desert, mountain, canyon, ocean and other complicated or remote places, it is difficult to achieve effective coverage in an economic way. It is estimated that in the 5G stage, 80% of the land and more than 95% of the ocean area in the world will still have no mobile signal [2]. This situation obviously can not meet the urgent needs of increasing broadband access for ocean navigation, scientific research and investigation, disaster rescue and other activities. The integrated space-aerial-terrestrial network can be a good solution to this problem.

At present, the industry has carried out a lot of research on the integration of satellite access and mobile network. In [3], the author proposes a lightweight service-oriented Space Air Ground Integrated Network (SAGIN) architecture. In [4] a SDN based network architecture that supports the integration of space, aerial and terrestrial access was proposed. The NGAT_SAT project [5] carried out by ITU-R proposed to integrate the satellite system into the future mobile network. ITU-T has carried out a series of research projects on fixed, mobile and satellite integrated access, and proposed multi access integrated network technology. 3GPP has carried out research on SAT_ARCH [6] and NTN [7] projects, committed to the combination of 5G network and satellite access. The proposed two converged network architectures including transparent elbow architecture and regenerative architecture as shown in Fig.1.

In the transparent elbow architecture, the satellite is completely transparent to New Radio (NR) signals including NR physical layer protocol. As a radio frequency remote unit, the satellite is responsible for transparently forwarding NR signals between User Equipment (UE) and gNB on the ground.

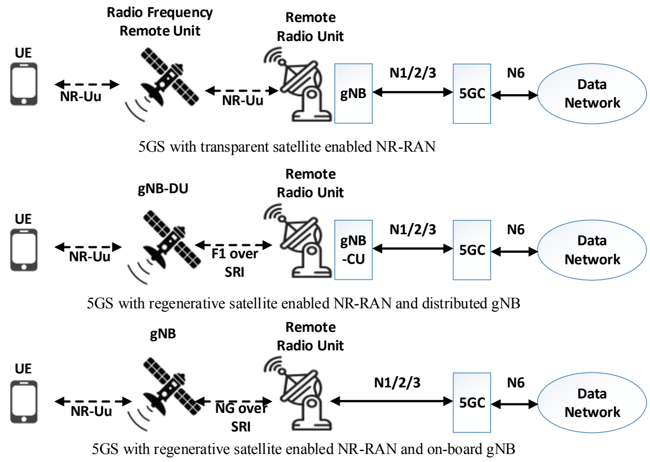

Fig. 1. Converged network architectures propsed by NTN project

In regenerative architectures. the Satellite can act as gNB-DU assumes part of the base station functions, and the F1 protocol between gNB-DU on Satellite and gNB-CU on ground is transmitted on Satellite Ridio Interface (SRI); The satellite can also undertakes all gNB functions, and the N1/N2/N3 interface protocols between gNB-DU on the satellite and 5GC on the ground are transmitted on SRI. In addition, satellites can be connected through Inter-Satellite Link (ISL) to achieve more flexible networking.

The satellites that can be integrated into the mobile network include LEO Satellite, Medium Earth Orbit (MEO) Satellite and Geosynchronous Orbit (GEO) Satellite. Compared with MEO/GEO Satellite, LEO Satellite has the outstanding advantages of low height, short transmission delay and less path loss, and can provide more efficient data distribution services for user terminals. Therefore, the development of LEO satellite has shown an explosive trend in recent years. Various organizations have launched the networking plan of constellation composed of multiple satellites with LEO satellite as the main body, Including Starlink, Hongyan, etc. High speed LEO satellite will frequently switch with terminals and base stations on the ground, which brings great challenges to the mobility management of mobile network. This paper focuses on the analysis of mobility management technology of converged network with satellite access.

This paper has the following contributions. Firstly, we analyze the advantages and disadvantages of various mainstream mobility management methods. Secondly, based on the comprehensive analysis of the current 5G mobility management methods, we put forward the optimization idea of mobility management for satellite access. Finally, two kinds of satellite access mobility management methods proposed by SAT_ARCH project are analyzed, and the future research direction of 6G network mobility management is proposed.

This paper is organized as follows. In chapter 2, the main methods of mobility management are introduced and analyzed. In the chapter 3, we comprehensively analyze the current 5G mobility management methods. In chapter 4, the optimization idea of mobility management method of satellite access is proposed. In Chapter 5, two mobility management methods proposed by 3GPP SAT_ARCH project are compared and analyzed. Chapter 6 is the conclusion.

II. EXISTING MOBILITY MANAGEMENT METHODS

A. *Mobile IP*

The design purpose of mobile IP is to keep the terminal IP address unchanged and ensure the continuity of data service after the mobile equipment changes the network location.

A.1 MIPv6

RFC 6275 [8] defines the basic framework of MIPv6. The MIPv6 framework mainly includes the following entities:

- MN: Mobile Node. MN is generally a host or router. It can move from one link of the network to another link, but it does not need to change the IP address or interrupt the ongoing communication;

- CN: Correspondent Node. Cn is the node communicating with MN;

- HA: Home Agent. HA is generally a router, which is responsible for maintaining the binding relationship between MN CoA and HoA, and transmitting the data sent to MN HoA address to MN CoA address through tunneling technology.

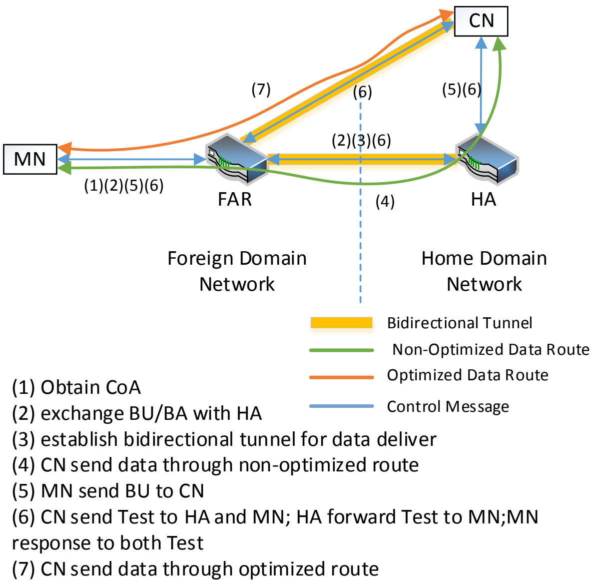

Fig. 2. MIPv6 schematic diagram

MN has two IP addresses in MIPv6 network.

- HoA: Home Address. The address is assigned by the HA of the home network and is used as the external communication address of MN;

- CoA: Care of Address. This address is the local address generated by MN in the foreign network according to the subnet mask of the access router. During the movement of MN, the HoA address remains unchanged, but the CoA will change with the location change. After the CoA changes, MN needs to send the new CoA to HA for rebinding.

When MN moves to the foreign domain network, it initiates a Binding Update (BU) request to HA, which carries the newly acquired CoA and HoA. Subsequently, a bidirectional tunnel will be established between HA and Foreign Access Router (FAR) for subsequent data forwarding between CN and MN. Because the data sent from CN to MN always goes through HA, there is a problem with circuitous route. MN can also send BU request directly to CN to establish an optimized data route, and then CN can communicate with MN directly through this optimized path.

A.2 HMIPv6

The HA is an anchor point for centralized mobility management. However, in a large network, MN bind to HA every time they change the CoA, which easily lead to HA signaling congestion. Therefore, IETF subsequently proposed a localized mobility management scheme HMIPv6 [9].

HMIPv6 introduces a new functional entity called Mobility Anchor Point (MAP), which is responsible for the local mobility management of MN. In HMIPv6, MN has two types of CoA addresses, one is Link Care of Address (LCoA). After MN obtains LCoA, it will use LCoA to bind to local MAP. The other is the Regional Care of Address (RCoA). After MN obtains the RCoA, it will use RCoA to bind and register with the HA of the home network. When MN switches between Access Routers (AR) within the scope of MAP service, MN only needs to use the new LCoA address to initiate BU to the current MAP instead of HA. Only when MN switches across MAPs, MN needs to use the new RCoA to initiate BU to HA.

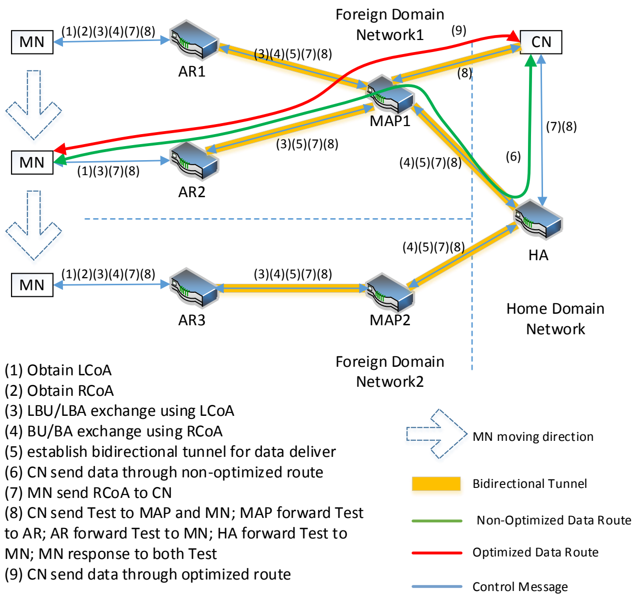

Fig. 3. HMIPv6 schematic diagram

When MN switches from AR1 to AR2, as it is still in the service area of MAP1, it only needs to use the newly obtained LCoA2 to initiate binding update to MAP1. When MN switches from AR2 to AR3, since it has changed to the server of MAP2, it not only uses the newly obtained LCoA3 to initiate binding update to MAP2, but also uses the newly obtained RCoA2 to initiate binding update to HA.

A.3 P-HMIPv6

MIPv6 and HMIPv6 are mainly used in legacy IP networks. If they are applied in mobile networks, MN's continuous movement will inevitably carry out BU operation frequently, resulting in problems such as excessive power consumption and excessive signaling. Therefore, on the basis of mobile IP protocol, IP paging [10] is introduced to strike a balance between accurate location information and low power consumption and signaling load.

P-HMIPv6 is a scheme that introduces paging mechanism based on HMIPv6. The mechanism has the following main characteristics:

- Allow MN to enter dormant state when there is no communication demand. Before entering the dormant state, MN will perform a local binding update to register its paging area with MAP. After entering the dormant state, MN will not initiate binding update as long as it is still moving in the same paging area.

- For MN in dormant state, MAP does not know its specific location, but only knows which paging area it is in.

- When MAP receives downlink data, if the CoA address bound by MN is not found, it will cache the downlink data first, and then broadcast paging messages to the current paging area of MN. When the target MN is found, and the MN changes to the active state and uses the CoA address for binding update, the MAP will send the cached data to the MN.

MIPv6 is the most widely used mobility management method in legacy IP networks. On the basis of the MIPv6, there are a lot of optimization methods. In addition to the HMIP described above, the hierarchical architecture is added to reduce binding signaling. P-HMIPv6 proposes a paging mechanism for dormant MN to save signaling interaction. FMIPv6 [11] is also used to reduce the delay and packet loss during handover, PMIPv6 [12] implements MIPv6 through the network side to reduce the requirements for terminals.

Mobile IP is the most widely used and longest used, so the device is well supported and the method is relatively mature. However, Mobile IP is still a centralized management method in essence. When users access, they need to access the centrally deployed anchor HA, which is difficult to solve such problems as circuitous data routing and poor scalability.

B. DMM

Distributed Mobility Management (DMM) [13] is proposed to solve the above problems.

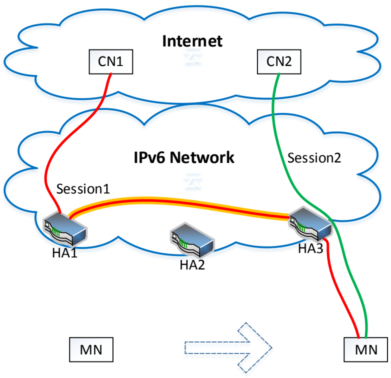

Fig. 4. An example of DMM

Fig.4 is a flow example of DMM. HA1, HA2 and HA3 in the figure are distributed in the network. They can be used as the anchor points of MN. MN can select HA access nearby to establish a better transmission path. When MN moves, the original communication between MN and CN1 is still forwarded via HA1 through the tunnel, while the access initiated by MN to CN2 is forwarded through the nearby HA3.

DMM does not subvert the idea of mobile IP in essence, but only optimizes its centralized management anchor. DMM proposes decentralized deployment of HA. The terminal can flexibly select and change anchor points as required, and can also transmit data concurrently through multiple anchor points, which greatly optimizes the data routing and enhances the scalability and flexibility of the network.

C. Location/Identifier Split

Under the mobile IP architecture, MN's IP address (HoA) is used not only as the identification of users, but also as the identification of location information. The dual role of IP address seriously hinders the mobility support of the existing network, and brings great restrictions to the scalability of the network.

In order to solve this problem, the HIP Architecture [14] proposed by IETF adds a new layer called host identity layer between IP layer and transport layer. HIP completely separates the two roles of host ID and location ID of the original IP address. The IP address is only used as the location ID of the host. In addition, the host ID is added to identify the hosts of both sides of communication. Before communication, the two parties need to shake hands four times using the host ID to confirm the location ID of the other party.

In addition, LISP (Locator/Identity Separation Protocol) [15] is also an important implementation based on this idea. LISP divides the address space into an End Identifier (EID) and a Routing Locator (RLOC). The EID is a topology-independent local host Identifier, and the RLOC is a global location Identifier. During communication, the sender needs to query its current RLOC in the mapping system according to the target EID, and then use the obtained RLOC to send the data to the target user. At present, there are also many implementation methods of mapping system, such as Lisp-DDT [15], Lisp-tree [16], etc.

D. Analysis of existing mobility management methods

This paper mainly introduces three kinds of mobility management methods: MIP, DMM and LISP. The three methods have different emphases.

MIPv6 mainly realizes the location management through the binding mechanism, and ensures the routing accessibility and service continuity in the process of MN movement. Afterward, HMIPv6, P-HMIPv6 and other methods are optimized on the basis of MIPv6 to reduce system signaling. MIP mechanism can be said to be the cornerstone of current mobility management, and various mobility management methods are optimized based on the core idea of MIP.

DMM focuses on optimizing the data transmission path of MN in the moving process, reducing the path detour and data transmission delay; LISP not only optimizes the data transmission path, but also enhances the scalability and flexibility of the network.

III. ANALYSIS OF 5G MOBILITY MANAGEMENT METHOD

The 5G system (5GS) actually makes use of the above mobility management methods comprehensively. The following is a detailed analysis.

A. Application of mobile IP in 5G analysis

Although 5GS does not directly adopt the protocol of mobile IP, its internal idea of GTP protocol is consistent with that of mobile IP.

A.1 MIPv6 application analysis

When UE establishes a session in 5GS, it will establish a GTP tunnel between gNB and User Plane Function (UPF) (if the session needs to pass through multiple UPFs, GTP tunnel also needs to be established between UPFs), and obtain the UE IP address. Unlike MIPv6, UE will only obtain one UE IP address. However, when establishing a GTP Tunnel, UE binds the IP address of the network elements at both ends of the Tunnel (gNB or UPF) and Tunnel Endpoint Identifier (TEID) to the UE IP address. This process is similar to the binding process of MIPv6, except that MIPv6 binds CoA address to HoA address. When the UPF receives the downlink data, it determines the GTP tunnel to which the user belongs according to the target IP address (i.e. the IP address of the UE), and sends the data packet to the peer end (gNB or UPF) through the GTP tunnel. The IP address and TEID of the peer end are carried in the tunnel header. The receiver can know the tunnel to which the data belongs according to the TEID of the tunnel header and deliver them in turn.

A.2 HMIPv6 application analysis

Similar to HMIPv6, GTP protocol is segmented, and the UE can move locally without rebinding the entire GTP tunnel.

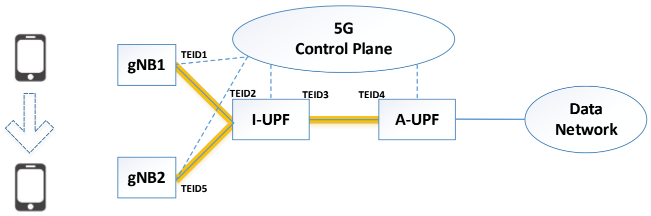

Fig. 5. GTP Tunnel modification

UE initially accesses 5G network from gNB1 and establishes a session. GTP tunnels are established between gNB1 and Intermediate UPF (I-UPF), and between I-UPF and Anchor UPF (A-UPF). Both ends of the tunnel between gNB1 and I-UPF are identified by TEID1 and TEID2 respectively, and both ends of the tunnel between I-UPF and A-UPF are identified by TEID3 and TEID4 respectively.

When the UE moves out of the service area of gNB1 and enters the service area of gNB2 during data transmission, the UE will immediately switch to gNB2 and dismantle the tunnel between gNB1 and I-UPF after establishing the GTP tunnel between gNB2 and I-UPF. In this process, the UE session does not need to update the GTP tunnel between I-UPF and A-UPF, which is similar to the binding between MAP and HA.

A.3 P-HMIPv6 application analysis

In 5GS, several base stations are divided into a Tracking Area (TA), and several TA are combined into a TA list. When the UE registers to access the 5GS, Access and Mobility Management Function (AMF) will assign a TA list to the UE. When the dormant UE is moving within the TA list, it is no need for the UE to update the location information to AMF. Only when the UE moves out of the current TA list area will the location update operation be carried out to the AMF, and the AMF will also assign a new TA list to the UE.

When there is downlink data to be sent to the dormant UE, UPF will cache the downlink data and notify AMF to page the user via SMF, and AMF will page the user within the current TA list range of the UE. After the user responds to the paging and enters the active state and to re-establishes the air link, the UPF will send the data to the UE.

It can be seen that AMF is similar to the control plane of MAP and is responsible for the management of the current paging area of dormant UE. The current TA list of UE in 5GS is equivalent to the paging area in P-HMIPv6.

B. DMM application analysis

Aiming at the problem of centralized access management of legacy 3G and 4G networks, 5GS introduces the concept of Session and Service Continuity (SSC) mode to enhance the mobility and continuity of 5G sessions. 5GS defines three SSC modes [17]. SSC Mode3 adopts the idea of DMM.

In SSC Mode3, the UE can select the nearest UPF to create a new session during moving, and the new service can access the network nearby through the new UPF. Original service is routed back to the original anchor point through the tunnel between UPFs. It is easy to see that SSC Mode3 actually uses the idea of DMM. While users can choose and change the anchor point to access nearby at any time, meanwhile it ensures the continuity of the original service.

C. Location/Identifier Split application analysis

In 5GS, IP address is not used to identify users or terminals, but only for data routing. After the downlink data is routed to the Anchor UPF according to the UE IP, the Anchor UPF will determine its GTP tunnel according to the UE IP address, and then use TEID for tunnel transmission.

If the UE is in dormant state, AMF will page the user in the user's current paging area, which carries the Temporary Mobile Subscriber Identity (TMSI) which is a user identity, and obtain the user's current specific location information (i.e. the current base station) according to the paging response. This paging process of finding the user's base station through the TMSI is similar to the process of querying the user's location information according to the EID mapping system in LISP.

IV. OPTIMIZATION IDEAS

At present, 5G mobility management mechanism has been able to meet various needs of ground mobile communication network, but the situation will change significantly after the introduction of satellite access. When studying the mobility management method of satellite access, the following factors need to be paid attention to:

1) Movement speed.

Unlike GEO satellite, LEO satellite always revolves around the earth at high speed. One beam of LEO satellite only lasts for a few minutes for geostationary users. This means that when users communicate through satellite access, they must switch every few minutes. On the one hand, it will bring a lot of switching signaling, on the other hand, it will also speed up the power consumption of the terminal.

2) Coverage.

Compared with the base station of the legacy network, the ground coverage of LEO satellite is much larger. Take the typical LEO satellite with 600km altitude and 30 ° inclination angle as an example, its ground coverage area is about 0.45% of the earth's surface area [9]. With such a large coverage, the number of users that a satellite can serve at the same time will be very considerable. Combined with the fact that users need to switch every few minutes, it is easy to form a signaling storm.

3) Propagation delay.

Satellite communication generally has large transmission delay. Take LEO satellite as an example. According to the difference of altitude and inclination angle, the one-way propagation delay of air interface ranges from 2.7ms to 14.8ms [6]. In the most extreme case, one-way air interface propagation of GEO satellite is even as high as 138.9ms. Unreasonable path selection or planning is likely to lead to an unacceptable end-to-end time delay. As shown in Fig.6, in the transparent elbow mode, the communication between two users under the same satellite needs to pass through the anchor point on the ground, resulting in four times satellite air interface propagation, which greatly increases the end-to-end delay.

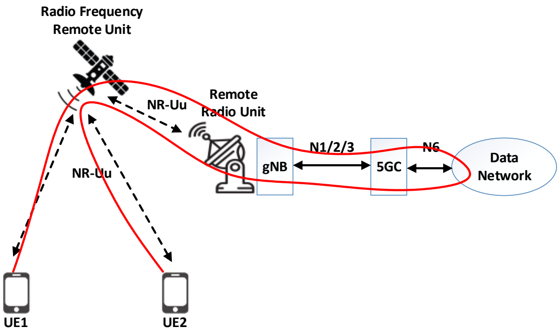

Fig. 6. Example of Non-Optimized Route

Considering the key factors of the above analysis, we propose two main optimization directions for mobility management of converged networks supporting satellite access.

1) Reduce the management cost

According to the legacy location management method, the high-speed movement of satellites will inevitably lead to the replacement of service satellites by users, resulting in a large number of frequent location update signaling. Therefore, how to avoid frequent signaling interaction as much as possible or reduce the impact of frequent signaling interaction on the system as far as possible is an important optimization direction.

Further, analysis and optimization can be performed in the following two different scenarios.

When the UE is in the active state for communicating, in order to maintain service continuity, the UE must immediately switch the service satellite after the service satellite moves out of the service range. The primary goal of this scenario is to ensure the user's service experience. Therefore, the frequent signaling interaction caused by the rapid movement of the satellite must be borne. At this time, the focus of optimization is how to select a better target satellite, so as to ensure that the target satellite can serve the user for as long as possible, thus reducing the frequency of satellite switching in disguised form. For example, the satellite with the moving direction as consistent as possible can be selected in combination with the moving trajectory of the satellite and the user.

When the UE is in dormant state, because there is no ongoing service, there is no need to record the user's location information very accurately and in real time. Various flexible methods can be considered to reduce signaling interaction. For example, in the case of actively reducing the location update frequency, the current location of users can be predicted by combining artificial intelligence and past historical data to minimize the broadcast range, and then gradually expand the paging range after the prediction fails. It can also be considered to realize the location update of batch users in a signaling interaction process through signaling optimization.

2) Optimize the transmission path.

From the above propagation delay factors, it can be seen that the propagation delay of satellite air interface is much greater than that of ground communication. Therefore, to reduce the end-to-end delay of satellite communication, the key is to reduce the air interface transmission. If the communication anchor point is on the ground, even if two users are under the same satellite, as shown in Fig.6, their end-to-end communication needs to go through four sections of air interface transmission. Therefore, it can be considered to put some switching equipment on the satellite, so that the satellite access users can exchange data directly on the satellite, and even communicate through the Inter Satellite Link (ISL).

V. ANALYSIS ON MOBILITY MANAGEMENT METHOD SUPPORTING SATELLITE ACCESS

According to the mobility management optimization idea proposed above, we will analyze the two main mobility management methods proposed by 3GPP in SAT_ARCH [6] project.

A. Distributed gNB

In this scheme, the ground station and the satellites in the satellite constellation constitute a dynamic virtual distributed base station, which includes:

- The Non-GeoStationary Orbit (NGSO) satellite serving the specific TA region through beam steering and beam forming. In this distributed base station, the NGSO satellite changes dynamically. With the change of satellite position, the NGSO satellite moved out of the specific TA will be removed, and the base station will be composed of the NGSO satellite and ground station newly moved into the TA region.

- Ground station

- Inter satellite link

- Feeder link between ground station and NGSO satellite

This method follows the management based on the static TA in the legacy mobile network. When the user accesses the base station through the satellite through the NR Uu Interface, the satellite is completely transparent to the user.

This scheme consists of NGSO constellation, ground station and ISL to form a distributed gNB, which serves the geostationary TA. This scheme has no impact on the existing process of UE and 5G core network, but requires the satellite to have the functions of beam steering and beam forming to ensure that the satellite can stably provide services to the ground static TA area in the process of moving.

The distributed gNB mode can enable the existing 5G network to support satellite integrated access at the least cost, and its mobility management scheme is completely followed by the current 5G system.

B. Idle state mobility management based on binding TA

Different from the distributed gNB scheme, this scheme is based on binding TA, that is, each satellite undertakes gNB function and binds specific TA fixedly, instead of dividing TA based on static regions on the ground. In this case, even when the user is stationary, due to the rapid movement of NGSO satellite, the user will update TA frequently. As shown in Fig.7, the UE starts to access 5G Core (5GC) via gNB1 bound to TA1. With the removal of gNB1, the UE changes to the gNB2 service area bound to TA2. Due to the change of TA, the UE needs to update the TA even if it is stationary in the dormant state.

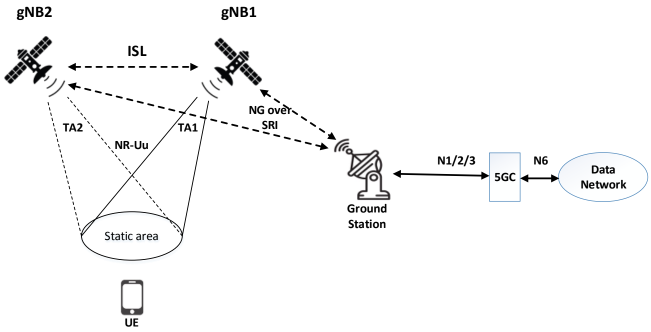

Fig. 7. NGSO satellite with binding TA

When UE initiates TA update from satellite access, AMF will return a TA list containing a group of sequential TAs to UE according to the current location of UE and the pre-configured trajectory of each NGSO satellite. The first TA is the TA of the satellite currently serving the user, and the subsequent TAs are the TA of multiple satellites that may serve the user of the location in order in the future according to the trajectory of the satellite. As shown in Fig.7, at the beginning, the user's activation Registration Area (RA) is TA1. With the gNB1 on the satellite leaving the user's area and the gNB2 on the satellite entering the user's area, the user finds that he has entered the service scope of TA2. At this time, the user changes the active RA to TA2. Since TA2 is just after TA1 in the TA list returned by AMF, it can be considered that UE is still within the predicted location range of the network, so UE does not need to actively update the location. When the TA of the satellite replaced by UE does not appear on the TA list in order, UE shall update the position immediately.

When AMF generates the TA list for UE, it will also generate a set of timers according to the speed and trajectory of each satellite in the TA list. When the timer of the first TA (such as TA1) times out, AMF will change the user's active RA to Subsequent TA and start the timer corresponding to it. AMF uses this method to implicitly maintain the activation RA of UE.

This method provides a useful exploration for satellite access, especially how to reduce the signaling interaction of mobility management as much as possible in the scenario of supporting on-board processing. However, this scheme is not a complete mobility management scheme. For example, it does not consider how to optimize the active scenario and how to reduce the detour of data transmission.

According to the above introduction and analysis of the two main mobility management methods in SAT_ARCH project, we can see that 3GPP currently only regards satellite as a supplement to 5G wireless access mode, and fails to optimize the system method according to the characteristics of NGSO satellite. In particular, the capabilities of satellite Internet and ISL were not fully utilized to optimize routing of data planes. In the future, 6G network should not only treat satellite access as a simple way of access, but also organically combine satellite Internet and mobile communication system, give full play to their respective strengths.

VI. CONCLUSIONS

The integration of space, aerial and terrestrial access is an important feature of 6G network in the future. Satellite access and satellite Internet will inevitably play an important role in 6G system. At present, the NTN architecture of 3GPP mainly regards satellite as the access mode of mobile network, and fails to comprehensively consider the mobility management method of converged network supporting satellite access from two aspects: reducing the management cost caused by frequent handover and optimizing data routing. For the future 6G network design, it is suggested to make full use of the characteristics and advantages of Satellite Internet and ISL to realize a real integrated space-aerial-terrestrial network.

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